Psychology of Testing

Psychology of Testing

- Historically testing was viewed as showing the system meets its requirements
- This has evolved to a stage where testing is performed with the primary aim of finding faults rather than proving correctness. It is perceived as a destructive process
- Seeking to find failures (the right mindset) can be viewed as criticism of the product and/or its author
- But looking for failures is constructive!
  - Time can be saved
  - Risks reduced
  - Costs reduced
  - Skills improved

Psychology of Testing

The Testing Mindset

- It is important that the Objectives of testing are clearly understood as humans will moderate their behaviour accordingly (however sub-consciously):-
  - "If testing is showing the system meets its requirements then I will just produce tests that show this."
  - "If testing is aimed at finding faults then I will be measured on this so I will put effort into designing tests that are more likely to find faults."
- The Testing mindset is different from a Developer's

Goal Conflicts About Testing

- You (as developer) are blind for your own mistakes.
- You are constructive, testing is destructive.
- You want to get finished. Testing delays you.
  - Someone else should test!
- Learning takes time someone else.
  - Test yourself!
- A good test cannot save a bad program.

Psychology of Testing

Testing Pearl of Wisdom



- ***"Testing is an extremely creative and intellectually challenging task"***
- ***"Tests must be written for invalid and unexpected, as well as valid and expected, input conditions"***

Myers - 1979

Psychology of Testing

Traits of Good Testers

- A Tester needs:
  - good communication skills
  - good observation skills
  - people handling skills
  - Curiosity
  - patience
  - reliability
  - thoroughness
  - an inquisitive nature
  - attention to detail
  - creativity in terms of identifying likely faults
  - Experience





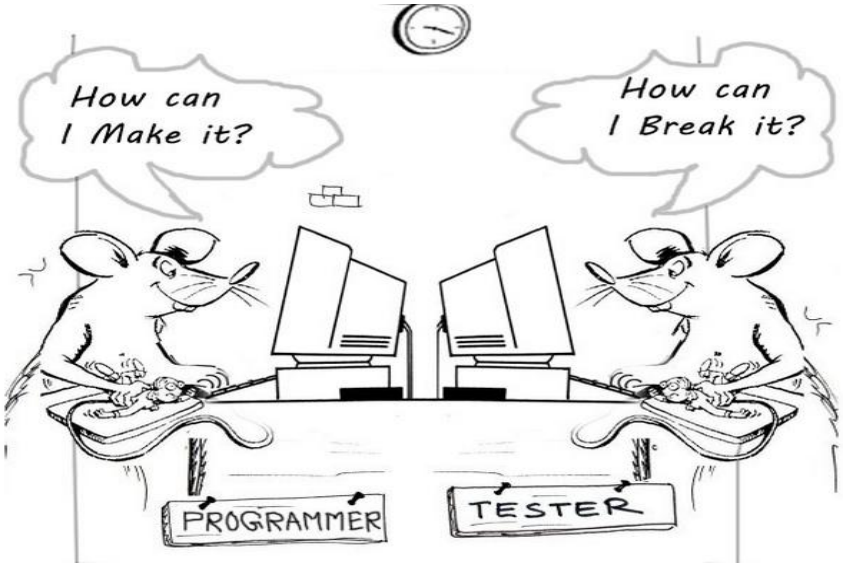
- However as with most other disciplines an effective test team will need a mix of skills so it is difficult to generalise

Psychology of Testing

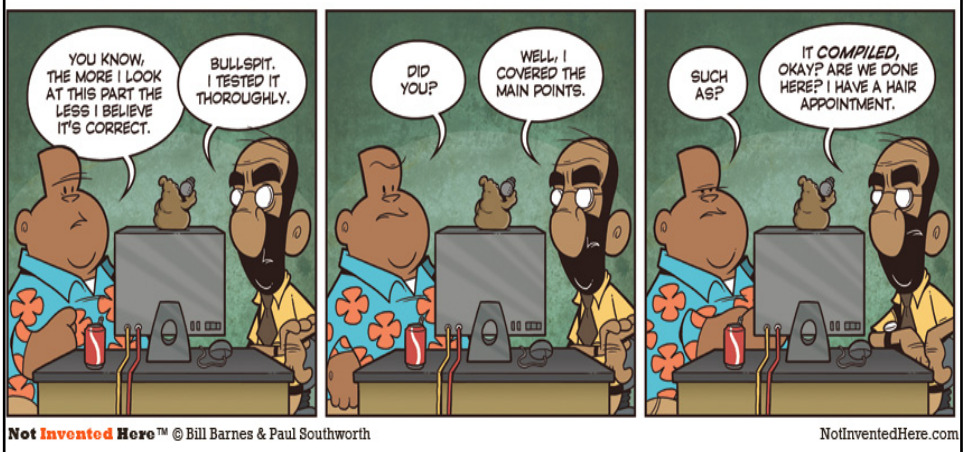
Developer vs Tester Relationship

- The relationship between a Developer and a Tester is not normally an easy one because:-
  - testers point out problems with software
  - developers like to think their software is perfect
  - testers are perceived as delaying the project by finding faults in the system
  - when the development slips testers normally have to work long hours to test the product, which in turn can cause resentment.
- It is important that they work together
- It is also important that they have mutual respect for each other.
- Collaboration is the right approach – we work to a common goal!
- Communicate findings objectively, not subjectively

Developer vs Tester Relationship



Developer vs Tester Relationship

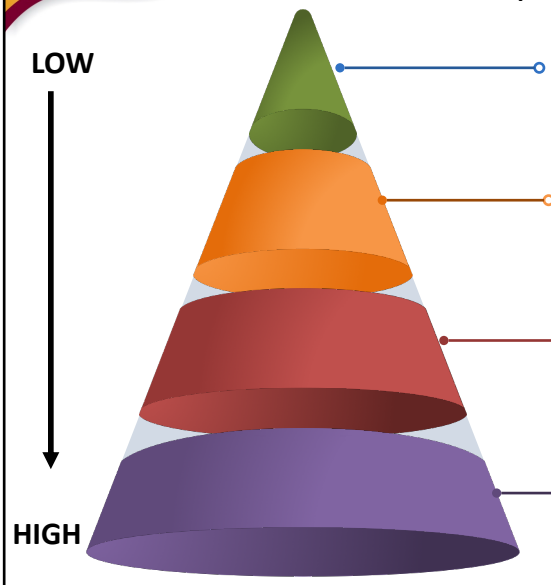


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Independent testing

- The right mindset could enable Developers to test the code
- However, passing this responsibility to trained and professional testing resources has many benefits (such as higher defect find rates)
- Authors tend to bring across assumptions they have made when developing the software. They are less likely to write tests to show faults in their own software (human nature)
- With testing performed by independent testers, testing effort is focused and not compromised by development effort and bias
- It is generally believed that objective independent testing is more effective

Levels of Independence



Tests designed by the person(s) who wrote the software under test

Tests designed by another person(s) (e.g. from the development team).

Tests designed by a person(s) from a different organizational group (e.g. an independent test team)

Tests designed by a person(s) from a different organization or company (e.g. outsourcing to an in-house or external test specialist organisation)